

(1) Find local extrema and saddle points of the functions

(i) $f(x, y) = x^4 + y^4 - 4xy + 2$

(ii) $f(x, y) = x \sin y$

(iii) $f(x, y) = x^2 y e^{-x^2 - y^2}$

(2) Find the points on the surface $z^2 = xy + 1$ that are closest to the origin.

(3) Find the three positive numbers whose sum is 100 and whose product is maximal.

(4) Find the dimensions of a rectangular box of maximal volume for which the sum of the lengths of its 12 edges is equal to c .

(5) A cardboard box without lid is to have a volume of $32,000 \text{ cm}^3$. Find the dimensions that minimize the amount of cardboard used.

(6) If the length of the diagonal of a rectangular box is L , what is its largest possible volume?

(7) Calculate the iterated integrals:

(i) $\int_1^3 \int_0^1 (1 + 4xy) dx dy.$

(ii) $\int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x}\right) dx dy.$

(8) Calculate the double integrals:

(i) $\iint_R \frac{xy^2}{x^2 + 1} dA$, where $R = [0, 1] \times [-3, 3]$.

(ii) $\iint_R x \sin(x + y) dA$, where $R = [0, \frac{\pi}{6}] \times [0, \frac{\pi}{3}]$.

(9) Compute the volume of the solid that lies under the plane $3x + 2y + z = 12$ and above the rectangle $[0, 1] \times [-2, 3]$.

(10) Find the volume of the solid in the first octant bounded by the cylinder $z = 9 - y^2$ and the plane $x = 2$.

(11) Find the volume of the solid under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 9$.

(12) Evaluate $\int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} x^2 y^2 dx dy$.

(13) Compute $\int \int_D x \cos y dA$ where D is the region bounded by $y = 0$, $y = x^2$, $x = 0$ and $x = 1$.

(14) Find the volume of the solid between the surface $z = xy$ and the triangle in the plane $z = 0$ with vertices $(1, 1)$, $(4, 1)$ and $(1, 2)$.

(15) Compute $\int \int_D e^{-x^2-y^2} dA$ where D is the region bounded by the semicircle $x = \sqrt{4-y^2}$ and the y -axis.

(16) Find the volume of the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $y = z$, $x = 0$, $z = 0$ in the first octant.